# How do AI systems infer political identity? Evaluating AI sycophancy in chats about politics

**Introduction.** Research suggests that AI chat systems are sensitive to identity signals from user prompts. For example, researchers have used LLMs to estimate public opinion by prompting models with descriptions of survey respondent characteristics. These *synthetic surveying* techniques can estimate survey results with simulations of each respondent, next-token probability distributions, or probes of the model's internals [2, 6, 11]. When it comes to political questions, synthetic surveys can flatten the diversity of opinions held by partisans, better represent liberals than conservatives, and have less error for estimating the views of Democrats than Republicans [2, 11]. Meanwhile, research on *AI sycophancy* suggests that LLMs display social sycophancy by affirming the user's self-image [5]. When applied to political chats, LLM responses to election-related queries can be sensitive to demographic steering, especially by partisan affiliation, and political perspective sycophancy increases based on the model's understanding of a user's political views [4, 7]. Much of the past research on *synthetic surveys* and *AI sycophancy* has focused on explicit descriptions of user identity. But in practice, AI chat is most commonly used for practical guidance, writing, and information-seeking – domains where users are unlikely to disclose demographic information as explicitly as in simulated environments [3]. In real use cases, AI chatbots take cues from a wide range of topical areas to make inferences and generate output. Inspired by the upcoming 2028 U.S. general election and the growing prevalence of AI systems in American life, my proposed research focuses on how LLMs make implicit inferences in political chat contexts based on ostensibly non-political user interests. The proposed research will first investigate *how AI systems infer user partisan identity from interest cues* **(Phase 1)** and *the extent to which these inferences manifest as political perspective sycophancy* **(Phase 2)**. Then, my research will evaluate *how AI chat design interventions can support pluralism and political neutrality* **(Phase 3)**.

**Phase 1: Do LLMs make inferences about user identity based on interest cues?** AI sycophancy only occurs in interaction contexts where the model can infer the user's perspective [7]. Therefore, I will first investigate whether LLMs make consistent inferences about user identity based on expressions of user interests. Inspired by synthetic surveying methods, I will audit model inferences about users' partisan identity based on *user persona descriptions* as explicit interest cues and *synthetic interaction contexts* as implicit interest cues [1, 2]. Several user interests will be selected from areas with externally valid metrics for partisan interest. For example, Republicans are more likely than Democrats to participate in *hunting*, while Democrats are more likely than Republicans to believe in *astrology* [9, 10]. Likewise, listener data suggests that the U.S. audience of many musical artists has a partisan lean; Latin artist *Bad Bunny*'s audience tends to listen from Democratic-leaning locations, while country singer *Morgan Wallen*'s audience tends to listen from Republican-leaning locations [8]. Based on cues that indicate user interest in various topics, I will then prompt the LLM to infer the user's partisan identity with a survey question-style prompt (i.e., "Do you consider yourself a Republican, Democrat, an independent or something else?"). Using closed models, I can measure partisan identity inferences based on the model's output response choice; with open-weight models, I can more thoroughly audit the probability distribution of response choices and probe representations of user interests in model internals. This phase investigates Hypothesis 1: LLMs make inferences about partisan identity based on ostensibly non-political user interests. These inferences are robust, occurring across interest areas with both explicit and implicit cues.

**Phase 2: Do AI responses show signs of perspective sycophancy based on user interest cues?** Phase 1 establishes whether models make interest-based inferences about partisan identity; Phase 2 evaluates political AI sycophancy based on these inferences. First, I will use user interest cues validated in Phase 1 to seed model interaction contexts with interests that elicit partisan identity inferences. (If Phase 1 finds that LLMs *do not* infer partisanship from user interests in any context, Phase 2 will proceed with explicit partisanship cues instead.) Based on political questions asked in previous AI sycophancy studies [4, 5, 7], I will then audit AI responses to questions about political issues. This will include questions grounded in real human evaluations, such as from the PARETO and OpinionQA datasets [11, 13]. As in Phase 1, I will audit results on both closed and open-weight models to better understand how sycophancy manifests for

closed models, which have built-in safeguards for political information, and base versions of open-weight models that have not been trained on human feedback. Working with open-weight models will enable me to conduct auditing techniques that can pinpoint where inferences occur in the model architecture [6]. This will support ongoing work in mechanistic interpretability of AI systems and inform decisions about interventions proposed in Phase 3. Thus, I will investigate Hypothesis 2: User interest-based inferences about partisan identity lead to perspective sycophancy, in which LLM output about political issues affirms the user's inferred partisan identity.

**Phase 3: Can interventions for pluralistic alignment support political neutrality in AI chat?** Users who explicitly prompt AI models with their partisan identity may seek personalized political advice. However, AI sycophancy based on user interests is significant because users who *don't* disclose their political partisanship have a *reasonable expectation for political neutrality* in AI chat interactions. In the absence of explicit partisan identity cues, AI systems should be designed to serve people with different values and perspectives, also known as *pluralistic alignment* [12]. For example, if a user who indicates interest in Bad Bunny receives AI responses that affirm Democratic perspectives, the AI system has failed to account for the *variability in values and perspectives* among Bad Bunny fans. Following Sorensen *et al.*'s suggestions, Phase 3 will focus on enforcing pluralism in models to mitigate AI sycophancy in political queries [12]. Based on the PARETO dataset of AI responses about political issues that have been evaluated by people across the political spectrum [13], I will test prompt engineering and model fine-tuning interventions for generating responses that reflect a balanced perspective.

**Intellectual Merit**

The proposed research contributes to ongoing interdisciplinary efforts in political communication and human-AI interaction research. This work promotes AI safety and transparency by quantifying the implicit inferences that AI chatbots make about users and how those inferences shape political information on AI platforms. Code, data, and prompts used for the proposed research will be made public and accessible to other researchers who wish to replicate and build on this analysis. Additionally, the interventions proposed in Phase 3 could be further evaluated in studies with human subjects to measure real-world impact. Other extensions of this research could delve deeper into mechanisms of AI sycophancy in other high-impact domains, such as healthcare, legal advice, and personal finance.

**Broader Impacts**

Open research on human-AI interaction is imperative to keep up with the pace of AI innovation and adoption. This research supports recent U.S. government priorities to ensure neutrality in AI deployed in federal agencies and national security systems [14]. Further, auditing how AI algorithms shape political attitudes supports informed public discourse and healthy democratic institutions. Since the proposed research addresses these salient policy areas, I will lead outreach to audiences outside of academia. This could include engaging with journalists, the private sector, government, policymakers, and the public.

**References:** [1] L. Argyle et al. Out of one, many: Using language models to simulate human samples. *Polit. Anal.* [2] A. Chapekis et al., Can AI stand in for human survey-takers? Not really. *Pew Research Center*. [3] A. Chatterji et al., OpenAI Signals v2.0. *OpenAI* [4] S. Cen et al., Large-scale, longitudinal study of large language models during the 2024 US election season. *arXiv:2509.18446*. [5] M. Cheng et al., ELEPHANT: Measuring and understanding social sycophancy in LLMs. *Int. Conf. Learn. Represent*. [6] E. Jahanparast et al. What do large language models know about opinions? *Int. Conf. Learn. Represent.* [7] S. Jain et al., Interaction context often increases sycophancy in LLMs. *Proc. ACM CHI Conf. Hum. Factors Comput. Syst*. [8] J. Liaukonytė et al., The political leaning of music audiences. *Cornell Univ.* [9] K. Parker et al., America's complex relationship with guns. *Pew Research Center*. [10] C. Rotolo, 30% of Americans consult astrology, tarot cards or fortune tellers. *Pew Research Center*. [11] S. Santurkar et al., Whose opinions do language models reflect?, *Int. Conf. Mach. Learn.* [12] T. Sorensen et al., A roadmap to pluralistic alignment. *Int. Conf. Mach. Learn.* [13] J. Stray et al., Political neutrality as balanced approval: A large-scale human evaluation of AI responses. *arXiv:2605.28911*. [14] The White House, EO 14319: Preventing woke AI in the federal government.